Result 1 (restricted vector value). [Clean topology study, restricted vector value]
Let K be a type. Let *field* be a field structure on K. Let n, a and m be natural numbers. Assume $a + m \leq n$ (`bound`). Let d be a function from $\{0, \ldots, n-1\}$ to $\{\text{false}, \text{true}\}$. Assume subpath sources preserved$(n, a, m, \mathit{bound}, d)$ (`sources`). Assume subpath targets preserved$(n, a, m, \mathit{bound}, d)$ (`targets`). Let Z be a module diagram on Q_d. Let I be a type. Let L be a family of ordered interval labels indexed by I on vertices 0 through n. Let *vectors* be a function assigning to each $u : \{0, \ldots, n+1-1\}$ a value of type $(I \to Z.\text{object}(u))$. Let v be a vertex in $\{0, \ldots, m+1-1\}$. Let j be an element of $L|_{[a,a+m]}$. Then

$$\mathit{vectors}|_{[a,a+m]}(v)(j) = \mathit{vectors}(a+v)(j).$$

Result 2 (restricted vectors vanish). [Clean topology study, restricted vectors vanish]
Let K be a type. Let *field* be a field structure on K. Let n, a and m be natural numbers. Assume $a + m \leq n$ (`bound`). Let d be a function from $\{0, \ldots, n-1\}$ to $\{\text{false}, \text{true}\}$. Assume subpath sources preserved$(n, a, m, \mathit{bound}, d)$ (`sources`). Assume subpath targets preserved$(n, a, m, \mathit{bound}, d)$ (`targets`). Let Z be a module diagram on Q_d. Let I be a type. Let L be a family of ordered interval labels indexed by I on vertices 0 through n. Let *vectors* be a function assigning to each $u : \{0, \ldots, n+1-1\}$ a value of type $(I \to Z.\text{object}(u))$. Assume *vectors* vanish outside the intervals L (`vanishes`). Then

$$\mathit{vectors}|_{[a,a+m]} \text{ vanish outside the intervals } L|_{[a,a+m]}.$$

Result 3 (restricted vectors compatible). [Clean topology study, restricted vectors compatible]
Let K be a type. Let *field* be a field structure on K. Let n, a and m be natural numbers. Assume $a + m \leq n$ (`bound`). Let d be a function from $\{0, \ldots, n-1\}$ to $\{\text{false}, \text{true}\}$. Assume subpath sources preserved$(n, a, m, \mathit{bound}, d)$ (`sources`). Assume subpath targets preserved$(n, a, m, \mathit{bound}, d)$ (`targets`). Let Z be a module diagram on Q_d. Let I be a type. Let L be a family of ordered interval labels indexed by I on vertices 0 through n. Let *vectors* be a function assigning to each $u : \{0, \ldots, n+1-1\}$ a value of type $(I \to Z.\text{object}(u))$. Assume *vectors* respect the directed edges of Z on the source support of L (`compatible`). Then

$$\mathit{vectors}|_{[a,a+m]} \text{ respect the directed edges of } Z|_{[a,a+m]} \text{ on the source support of } L|_{[a,a+m]}.$$

LLM-assisted statements and proofs (gpt-6, 2026-11-07). See PROVENANCE.md. The vector evaluation, vanishing, and edge-compatibility statements are checked. The paper's pyramid theorem remains a separate proof obligation.